Problem-Solution Fit

Date	26/06/2026
Team ID	Not Assigned
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS]

• Meteorologists monitoring flood-prone districts • Disaster relief coordinators managing monsoon response • Government analysts assessing flood preparedness • Local authorities responsible for public safety • Flood risk researchers studying rainfall in India

6. CUSTOMER LIMITATIONS [CL]

• No access to real-time intelligent prediction tools • Budget constraints — cannot afford costly systems • Rely on desktop/laptop browsers only • Low technical expertise — need simple UI • Dependent on slow manual data collection

5. AVAILABLE SOLUTIONS — PROS & CONS [AS]

PROS: • IMD weather forecasting — widely trusted • Traditional models — established methods CONS: • Delayed alerts — not actionable in time • No web-based interface for field officers • Cannot process multiple features at once • No ML-based local district prediction

2. PROBLEMS / PAINS + FREQUENCY [PR]

• Flood warnings issued too late — Every monsoon (HIGH) • Manual rainfall analysis is slow — Daily (HIGH) • No instant flood risk assessment tool — Year-round (HIGH) • Resources deployed to wrong regions — Every event (HIGH) • No data-driven support for decisions — Ongoing (MEDIUM)

9. PROBLEM ROOT / CAUSE [RC]

• No automated ML-based flood prediction system • Over-reliance on traditional threshold models • No integration of historical data with real-time inputs • Lack of web-based interfaces for field officers • Insufficient use of data science in disaster management

7. BEHAVIOR + INTENSITY [BE]

• Meteorologists manually check rainfall thresholds — Daily (HIGH) • Coordinators call district offices for status — During floods (HIGH) • Analysts compile Excel weather sheets — Weekly (MEDIUM) • Authorities issue alerts from experience alone — Every event (HIGH) • Researchers manually analyse raw CSV data — Monthly (MEDIUM)

3. TRIGGERS TO ACT [TR]

• Heavy rainfall forecast exceeds historical averages • River water levels approach danger marks • Monsoon season begins (June–September annually) • Previous flood events causing community displacement • Government directives for improved disaster preparedness

10. YOUR SOLUTION [SL]

Rising Waters — ML Flood Prediction System: • XGBoost model — 96.55% accuracy on historical Indian rainfall data • Flask web app: enter 10 weather inputs → instant flood prediction • 4 ML models compared: Decision Tree, Random Forest, KNN, XGBoost • Deployed on Render.com — accessible globally via browser • transform save ensures consistent real-time data preprocessing • Probability score + advisory actions on result pages

8. CHANNELS OF BEHAVIOR [CH]

ONLINE: • rising-waters-flood-prediction.onrender.com • GitHub repository for code and version control • Jupyter Notebook for model training OFFLINE: • Anaconda Navigator for local development • Command Prompt for running Flask app • Local browser at http://127.0.0.1:5000

4. EMOTIONS BEFORE / AFTER [EM]

BEFORE: • Anxious and underprepared during monsoon season • Frustrated by slow, unreliable manual methods • Overwhelmed managing multiple at-risk regions AFTER: • Confident with instant data-driven flood risk results • Relieved by clear advisory recommendations • Empowered to issue timely warnings effectively